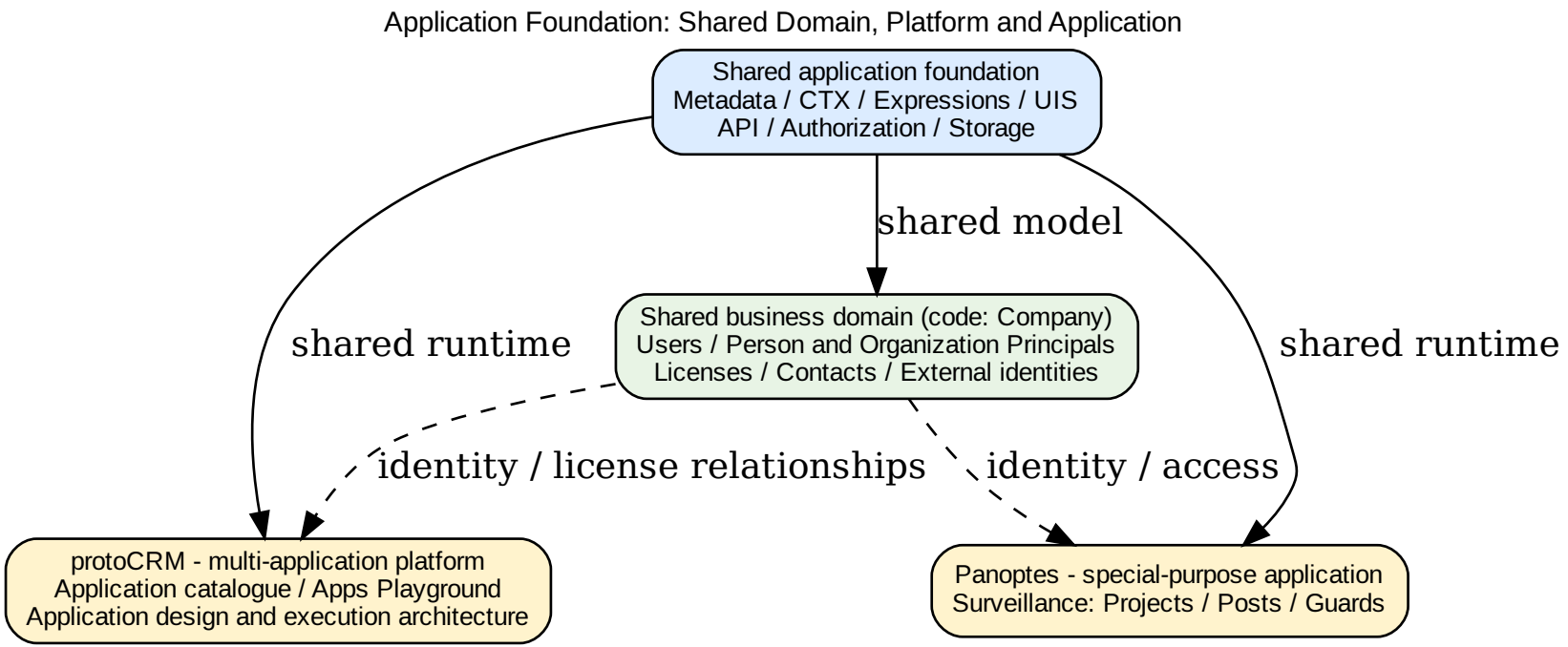
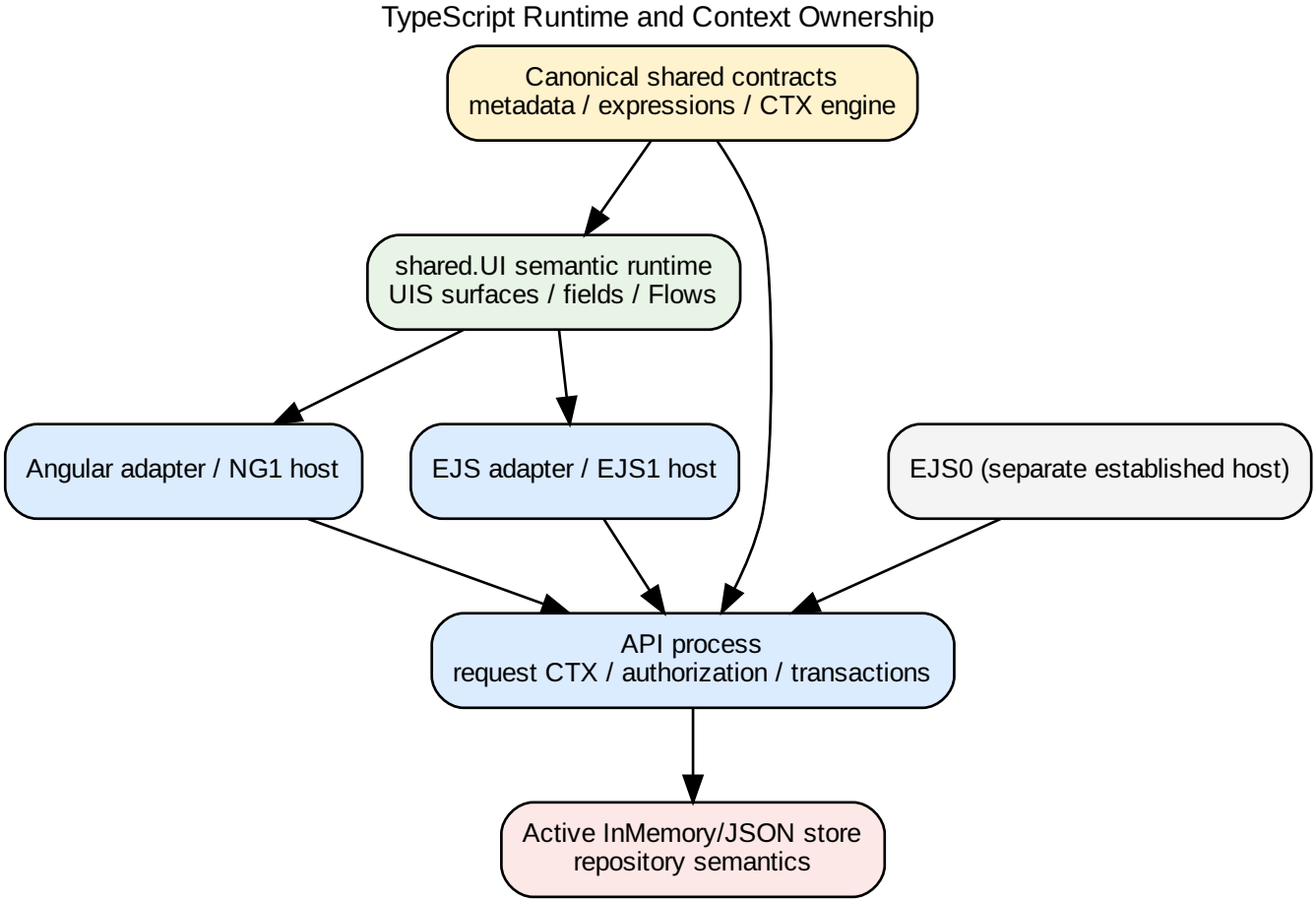
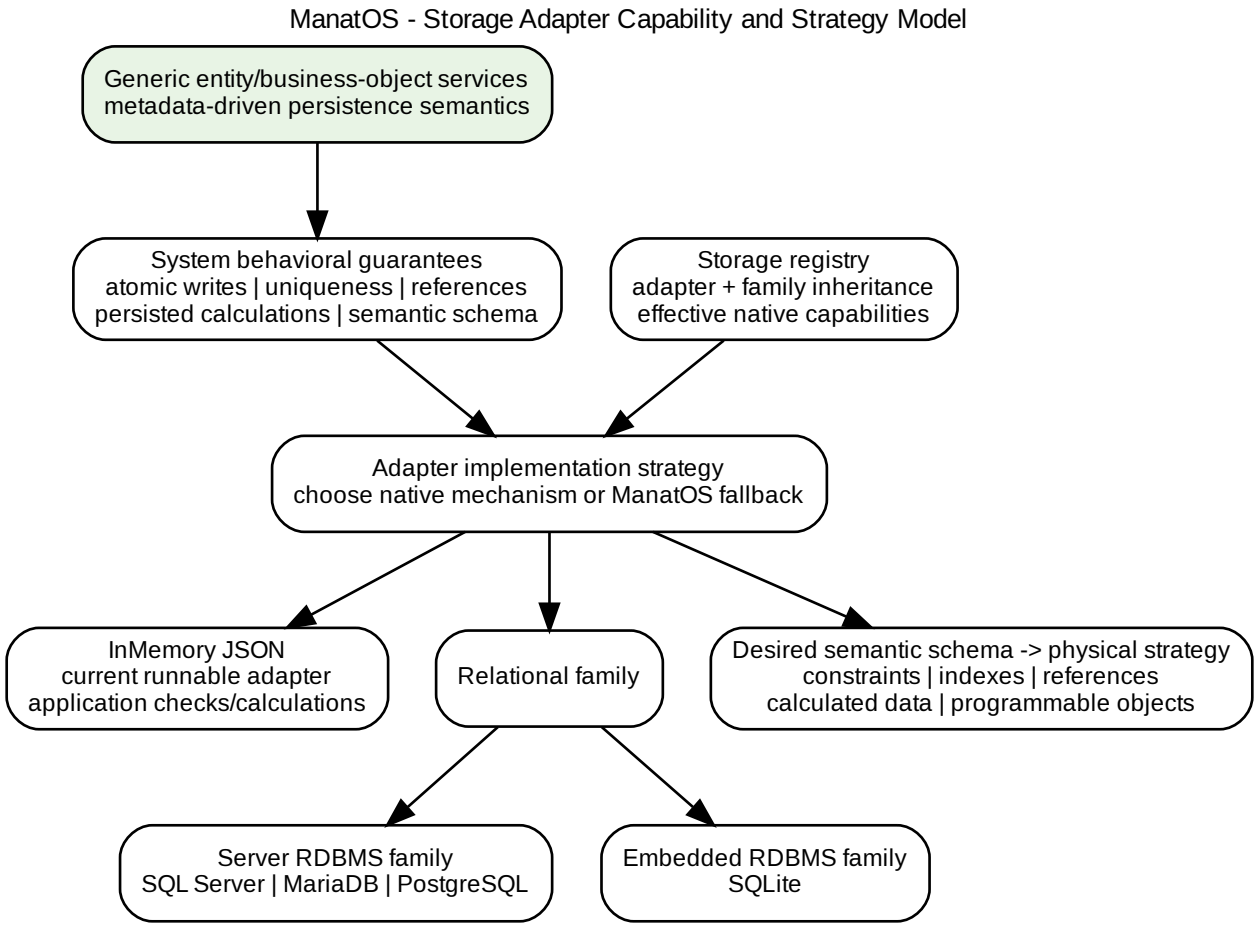
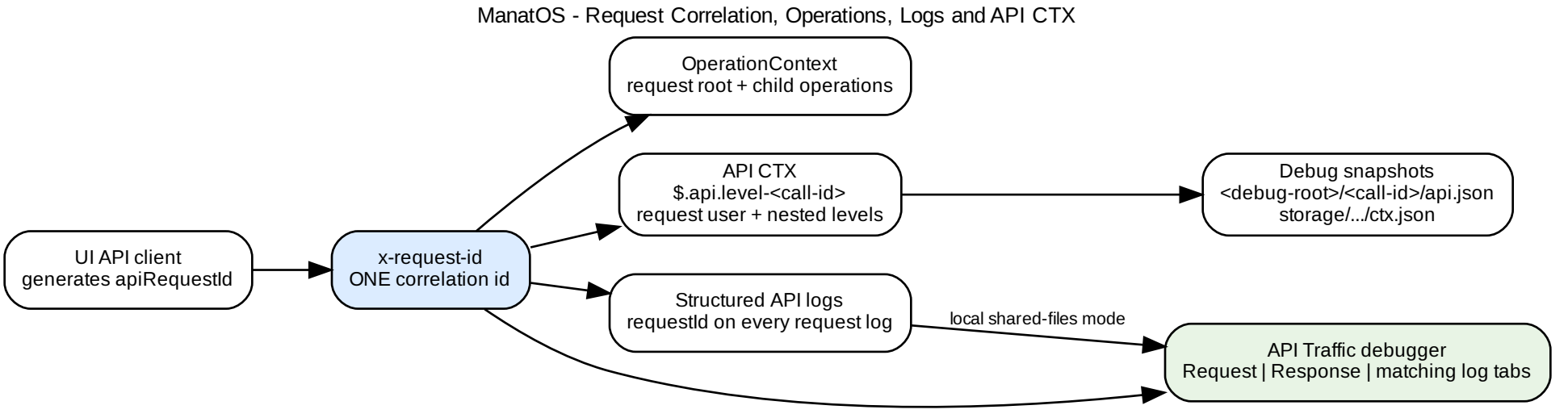
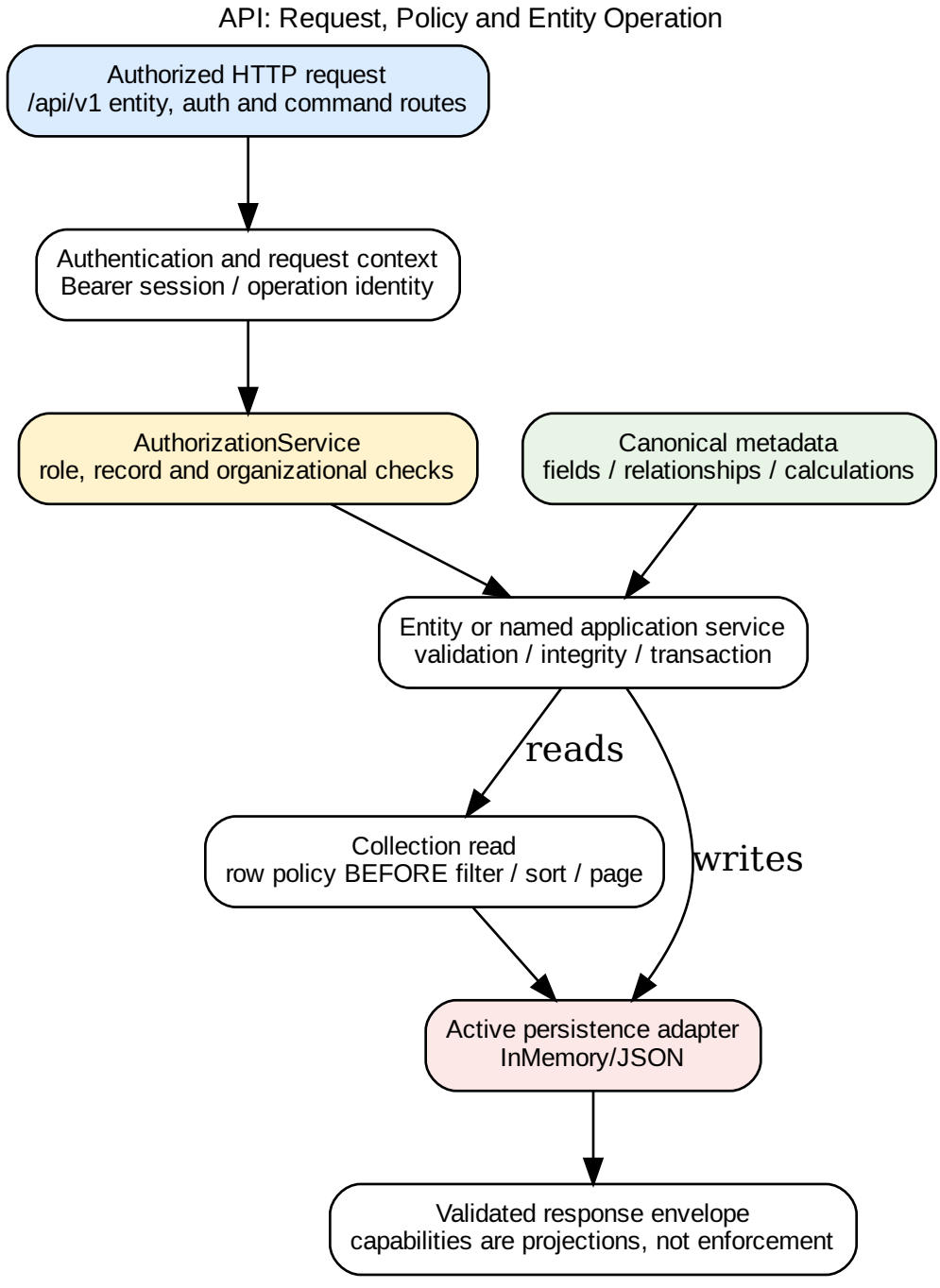
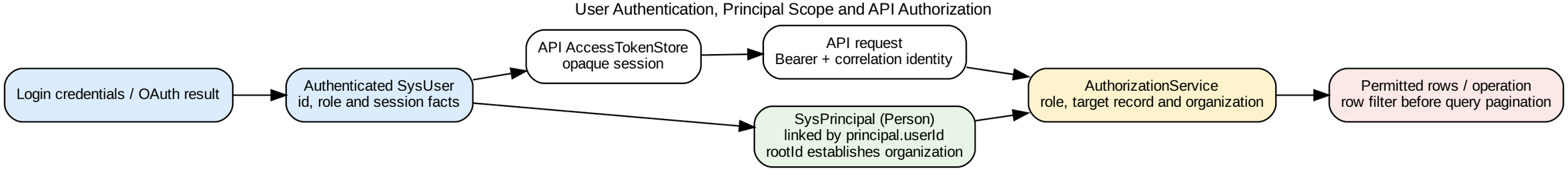

Application Foundation: Shared Domain, Platform and Application
Shared application foundation
Metadata / CTX / Expressions / UIS
API / Authorization / Storage
shared model
shared runtime
shared runtime
Shared business domain (code: Company)
Users / Person and Organization Principals
Licenses / Contacts / External identities
identity / license relationships
identity / access
protoCRM - multi-application platform
Application catalogue / Apps Playground
Application design and execution architecture
Panoptes - special-purpose application
Surveillance: Projects / Posts / Guards

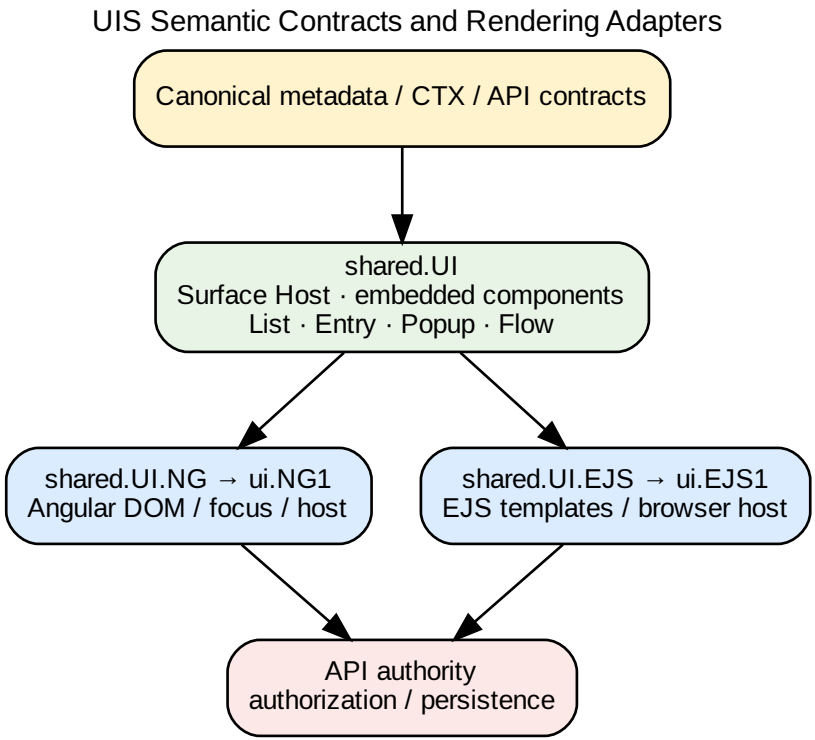

TypeScript Runtime and Context Ownership
Canonical shared contracts
metadata / expressions / CTX engine
shared.UI semantic runtime
UIS surfaces / fields / Flows
Angular adapter / NG1 host
EJS adapter / EJS1 host
EJS0 (separate established host)
API process
request CTX / authorization / transactions
Active InMemory/JSON store
repository semantics

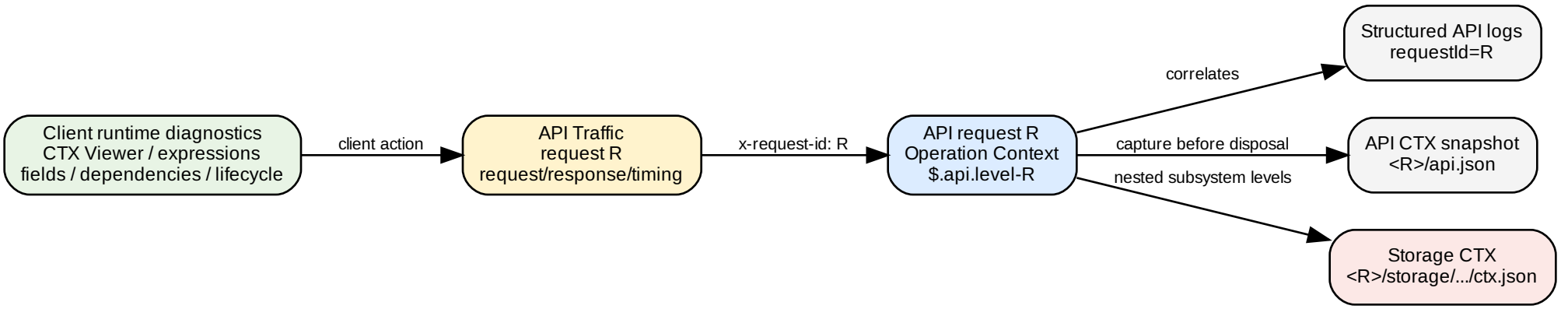

ManatOS - Storage Adapter Capability and Strategy Model
Generic entity/business-object services
metadata-driven persistence semantics
System behavioral guarantees
atomic writes | uniqueness | references
persisted calculations | semantic schema
Storage registry
adapter + family inheritance
effective native capabilities
Adapter implementation strategy
choose native mechanism or ManatOS fallback
InMemory JSON
current runnable adapter
application checks/calculations
Relational family
Desired semantic schema -> physical strategy
constraints | indexes | references
calculated data | programmable objects
Server RDBMS family
SQL Server | MariaDB | PostgreSQL
Embedded RDBMS family
SQLite

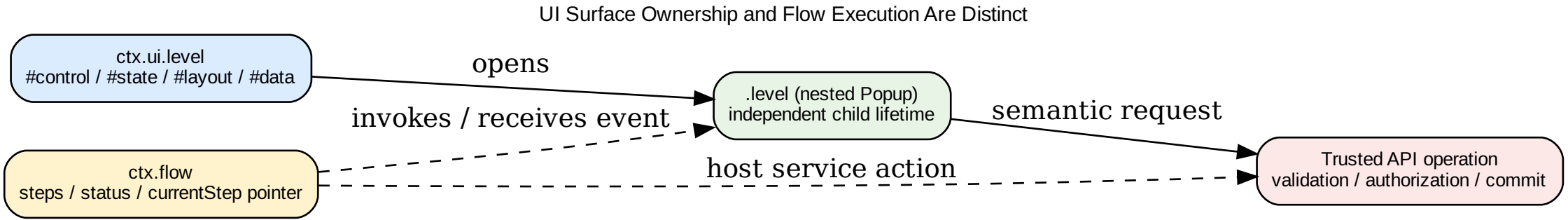

ManatOS - Request Correlation, Operations, Logs and API CTX
UI API client
generates apiRequestId
x-request-id
ONE correlation id
OperationContext
request root + child operations
API CTX
$.api.level-<call-id>
request user + nested levels
Structured API logs
requestId on every request log
Debug snapshots
<debug-root>/<call-id>/api.json
storage/.../ctx.json
local shared-files mode
API Traffic debugger
Request | Response | matching log tabs

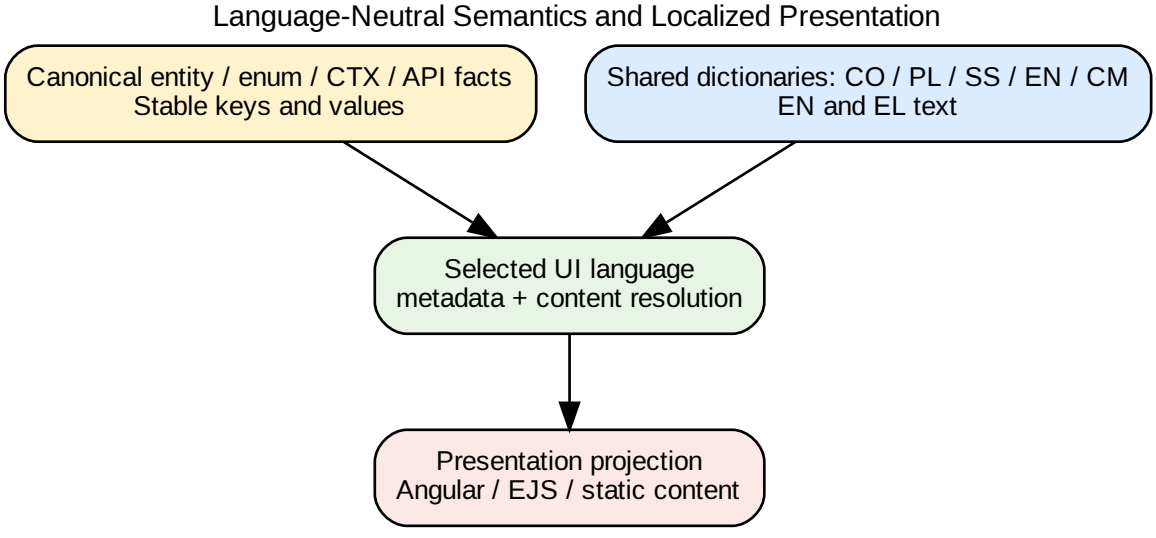

API: Request, Policy and Entity Operation
Authorized HTTP request
/api/v1 entity, auth and command routes
Authentication and request context
Bearer session / operation identity
AuthorizationService
role, record and organizational checks
Canonical metadata
fields / relationships / calculations
Entity or named application service
validation / integrity / transaction
reads
writes
Collection read
row policy BEFORE filter / sort / page
Active persistence adapter
InMemory/JSON
Validated response envelope
capabilities are projections, not enforcement

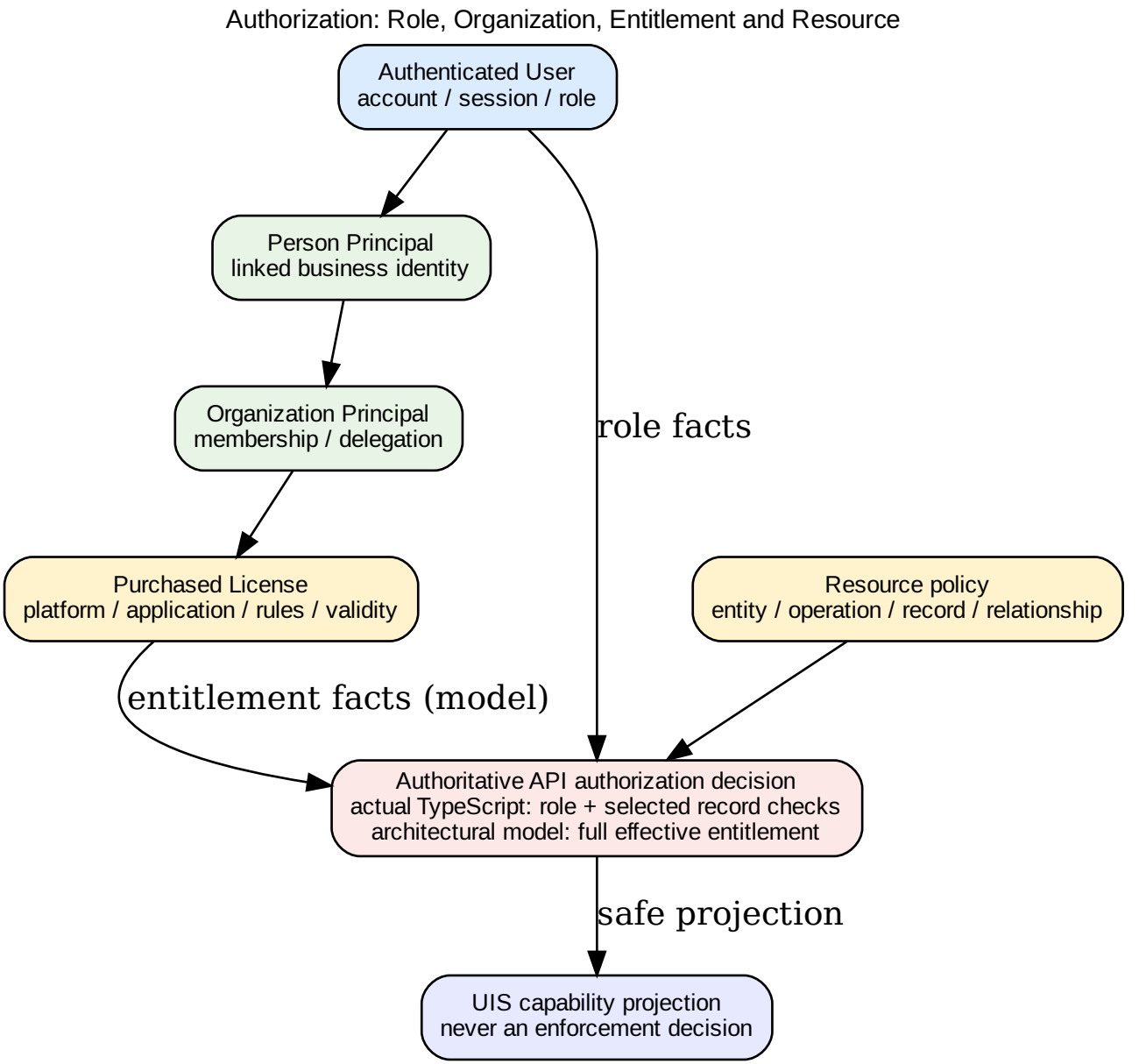

User Authentication, Principal Scope and API Authorization
Login credentials / OAuth result
Authenticated SysUser
id, role and session facts
API AccessTokenStore
opaque session
API request
Bearer + correlation identity
SysPrincipal (Person)
linked by principal.userId
rootId establishes organization
AuthorizationService
role, target record and organization
Permitted rows / operation
row filter before query pagination

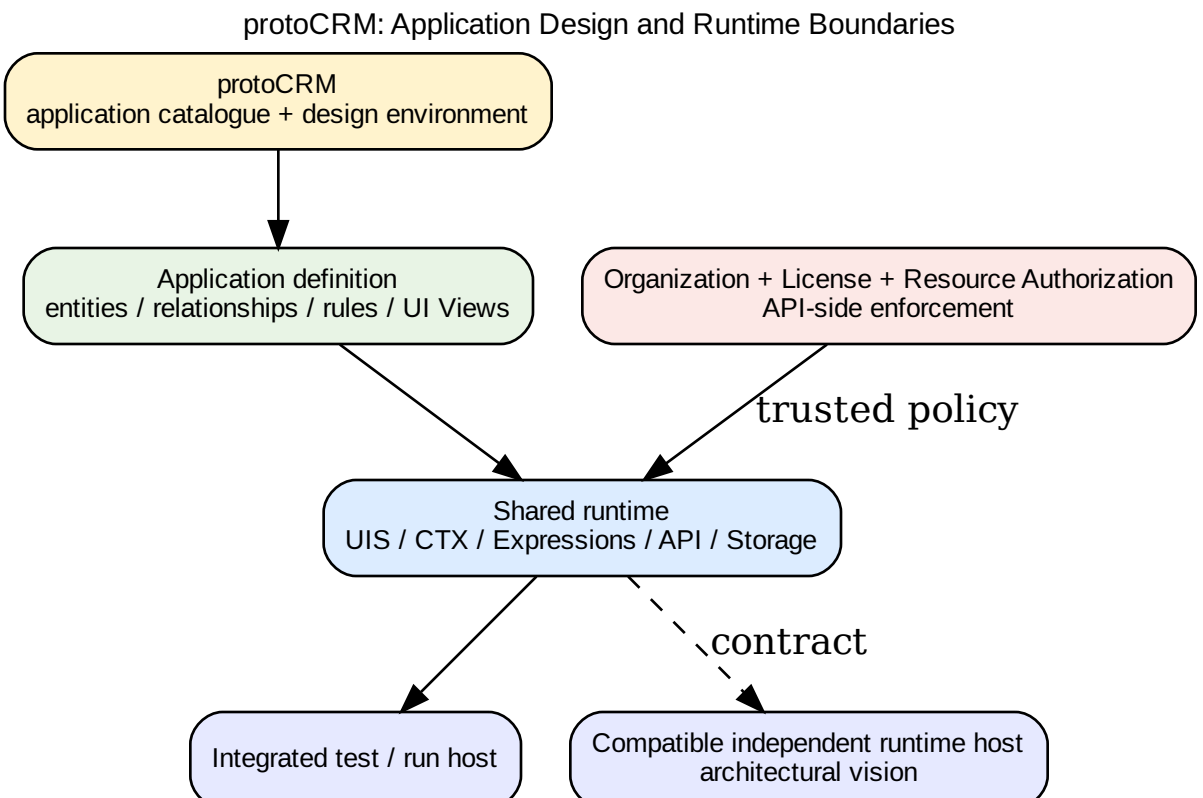

UIS Semantic Contracts and Rendering Adapters
Canonical metadata / CTX / API contracts
shared.UI
Surface Host · embedded components
List · Entry · Popup · Flow
shared.UI.NG → ui.NG1
Angular DOM / focus / host
shared.UI.EJS → ui.EJS1
EJS templates / browser host
API authority
authorization / persistence


ManatOS - End-to-End Diagnostics and Observability
Client runtime diagnostics
CTX Viewer / expressions
fields / dependencies / lifecycle
client action
API Traffic
request R
request/response/timing
x-request-id: R
API request R
Operation Context
$.api.level-R
correlates
Structured API logs
requestId=R
capture before disposal
API CTX snapshot
<R>/api.json
nested subsystem levels
Storage CTX
<R>/storage/.../ctx.json


UI Surface Ownership and Flow Execution Are Distinct
ctx.ui.level
#control / #state / #layout / #data
opens
.level (nested Popup)
independent child lifetime
semantic request
invokes / receives event
ctx.flow
steps / status / currentStep pointer
host service action
Trusted API operation
validation / authorization / commit


Language-Neutral Semantics and Localized Presentation
Canonical entity / enum / CTX / API facts
Stable keys and values
Shared dictionaries: CO / PL / SS / EN / CM
EN and EL text
Selected UI language
metadata + content resolution
Presentation projection
Angular / EJS / static content


Authorization: Role, Organization, Entitlement and Resource
Authenticated User
account / session / role
Person Principal
linked business identity
Organization Principal
membership / delegation
role facts
Purchased License
platform / application / rules / validity
Resource policy
entity / operation / record / relationship
entitlement facts (model)
Authoritative API authorization decision
actual TypeScript: role + selected record checks
architectural model: full effective entitlement
safe projection
UIS capability projection
never an enforcement decision


protoCRM: Application Design and Runtime Boundaries
protoCRM
application catalogue + design environment
Application definition
entities / relationships / rules / UI Views
Organization + License + Resource Authorization
API-side enforcement
trusted policy
Shared runtime
UIS / CTX / Expressions / API / Storage
contract
Integrated test / run host
Compatible independent runtime host
architectural vision